

Fish Consumption, Dietary Long-Chain n-3 Fatty Acids, and Risk of Type 2 Diabetes



OBJECTIVE The evidence on the association between fish consumption, dietary long-chain n-3 fatty acids, and risk of type 2 diabetes is inconsistent. We therefore performed a systematic review and meta-analysis of the available prospective evidence. **RESEARCH DESIGN AND METHODS** Studies were identified by searching the PubMed and EMBASE databases through 15 December 2011 and by reviewing the reference lists of retrieved articles. Prospective studies were included if they reported relative risk (RR) estimates with 95% CIs for the association between fish consumption and/or dietary long-chain n-3 fatty acids and incidence of type 2 diabetes. A dose-response random-effects model was used to combine study-specific RRs. Potential sources of heterogeneity were explored by prespecified stratifications. **RESULTS** Sixteen studies involving 527,441 participants and 24,082 diabetes cases were included. Considerable statistical heterogeneity in the overall summary estimates was partly explained by geographical differences. For each serving per week increment in fish consumption, the RRs (95% CIs) of type 2 diabetes were 1.05 (1.02–1.09), 1.03 (0.96–1.11), and 0.98 (0.97–1.00) combining U.S., European, and Asian/Australian studies, respectively. For each 0.30 g per day increment in long-chain n-3 fatty acids, the corresponding summary estimates were 1.17 (1.09–1.26), 0.98 (0.70–1.37), and 0.90 (0.82–0.98). **CONCLUSIONS** Results from this meta-analysis indicate differences between geographical regions in observed associations of fish consumption and dietary intake of long-chain n-3 fatty acids with risk of type 2 diabetes. In consideration of the heterogeneous results, the relationship warrants further investigation. Meanwhile, current public health recommendations on fish consumption should be upheld unchanged.